modifications. Whenever I complete a set of changes, I always use Antiword to produce a plain *.txt* file. I then copy the *.doc* and the *.txt*, appending a sequential number to the name, in order to have a backup. So, for example, *cv.txt.13* was produced from *cv.doc.13* using Antiword. If I want to see what I changed between, say, *cv.doc.13* and *cv.doc.16*, I use the following command:

```
diff cv.txt.13 cv.txt.16
```

It's quick, lightweight and easy to read. The format of my resume may not be perfect, but the words are correct. Most agencies simply toss the document into some sort of database, so that they can search for certain capabilities. I'm sure the database is not the least bit interested in the format in which the document displays. Resumes are usually emailed, not printed. It's not that the format is so dreadful; it's just that things like tables may cross pages. But it's always going to be difficult to guarantee that sort of layout. A different printer can cause the layout to change. If I want it to be perfect, I create a PDF.

There is one issue with Antiword: it does not handle *.docx*. In the past, this was not a problem, but as more users migrate to more recent products from Microsoft, *.docx* is becoming more common. I have seen nothing to indicate that the author of Antiword is planning modifications to handle *.docx*. And that's not altogether surprising. The difference between *.doc* and *.docx* is not just a few extra features; it's a completely different format. A *.docx* file is a zip file containing several other files, many of them *.xml*. If you are interested in converting *.docx*, you might investigate *docx2txt* at *docx2txt.sourceforge.net*.

An alternative to Antiword is *catdoc* from *www.wagner.pp.ru/~vitus/software/catdoc/* but I can't offer an opinion, because I haven't tried it.

The reason I have not looked at *docx2txt* or *catdoc* is because I now use LibreOffice (Writer—as detailed earlier) for this sort of conversion. And, yes, LibreOffice handles *.doc* and *.docx* seamlessly. Here's an example of how to run it from the command line (or a script):

```
libreoffice --headless --convert-to txt:Text AttributeReview.docx
convert /tmp/AttributeXReview.docx -> /tmp/AttributeXReview.txt
using Text
```

It's just some extra typing compared to Antiword, but you could avoid that with an alias.

Spreadsheets

For a while, I used *xls2csv* (which comes from the same stable as *catdoc*). It worked fine until it came to dates. There are several date formats used internally by Microsoft Excel, and *xls2csv* was getting them wrong in the documents I was trying to convert. In a state of confusion, I stumbled on a different program called *xls2csv* at *search.cpan.org/~ken/xls2csv-1.07/script/xls2csv*. I haven't tried this one because of what I read at *vinayhacks.blogspot.com.au/2010/04/converting-xls-to-csv-on-linux.html*. What I use instead is LibreOffice (Calc, as detailed above). It has no problems with dates; and it also handles *xlsx*. Here's an example of how to run it from the command line (or a script):

```
libreoffice --headless --convert-to csv Access.xls
convert /tmp/Access.xls -> /tmp/Access.csv using Text - txt -
csv (StarCalc)
```

Possibly the main drawback with LibreOffice is that it has a heavy footprint compared to the other applications I have mentioned. But, these days, if your alternative is to run XP or a later offering from Microsoft, you need a machine that can carry the load. Further, if you plan to follow my general approach and run some of these tools in a VM, you are going to need a fairly beefy machine. My experience has been that performance is brilliant. At work, I have an Intel i7 (which I didn't pay for). That's the whole point: this is about Linux at work—software for the office.

Oracle databases

For interactive work, SQLDeveloper provides an environment that's consistent across Microsoft and Linux platforms. It has a helpful GUI that makes it very easy to use. Once you've mastered the arcane syntax required to connect to the database, the SQL needed to extract data is not too hard. The GUI provides a lot of help.

To run, this tool requires IcedTea *(icedtea.classpath.org)*, the OpenJDK development tools. Install these with *yum install java-1.6.0-openjdk-devel*. The URL for SQLDeveloper is *http://bit.ly/99P9Zv*. The total download size is 148 M. Next, install SQLDeveloper with *yum --nogpgcheck localinstall sqldeveloper-3.0.04.34-1.noarch.rpm*.

However, for unattended operation, I choose SQLPlus. Although there are a number of irritating things about it, I can ignore them all once I have extracted the data I need out of an Oracle database into a CSV. The URL is *www.oracle.com/technetwork/topics/linuxsoft-082809.html* for Instant Client Downloads for Linux x86. You can download *oracle-instantclient11.2-basic-11.2.0.3.0-1.i386.rpm and oracle-instantclient11.2-sqlplus-11.2.0.3.0-1.i386.rpm*. Install these with *yum --nogpgcheck localinstall oracle-instantclient11.2-basic-11.2.0.3.0-1.i386.rpm* (total size: 168 M) and *yum --nogpgcheck localinstall oracle-instantclient11.2-sqlplus-11.2.0.3.0-1.i386* (total size: 2.6 M).

Putting it all together

I'm going to outline the sort of tasks that I use these tools for. Typically, people maintain or generate some data, and then